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Phonogram-based Automatic Typo Correction in Malayalam Social Media Comments

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Abstract

The Internet has emerged as a pivotal medium for human interaction, leading to profound transformations in language dynamics, especially in computer-mediated communication (CMC) platforms. These transformations are reflected in changing orthography, graphology, vocabulary, grammar, syntax, pragmatics, and the style of natural languages. Internet-based communication has become more accessible. To increase accessibility, users of Internet-based communication platforms speak in the language that comes more naturally to them. Internet users use new terminologies and phrases to make their messages more engaging. Internet users all over the world are following this trend. Internese is the name of this emerging trend in internet lingo. All the internese features are vividly marked in Malayalam social media comments. Typographical errors are common in Malayalam social media comments. None of the spell checkers currently available in Malayalam recognize words and phrases in the social media comments scripted in Malayalam. This study proposes two different models of automatic typo detectors and typo correctors for Malayalam social media comments. The first model of Malayalam typo detectors and typo correctors is a sequence-2-sequence deep learning model, and the other is a hybrid model. These two models analyzed the input words at their phonogram level. The proposed hybrid system has achieved acceptable performance with the recently built corpus. The sequence-2-sequence model performs rather well. Because of Malayalam's intricate structure, the model's significant reliance on data influences its accuracy.

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Keywords: *Phonogram, Sequence-2-sequence model, graph model, Internet linguistics, typo, internese*

1. Introduction

The Internet, one of humanity's most remarkable inventions (Grigor Baghdasaryan, 2015), has fundamentally transformed how individuals interact, communicate, and access information. The advent of Computer-Mediated

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Communication (CMC) has profoundly impacted various aspects of daily life, facilitating swift and efficient exchange of information. Email, instant messaging, and social networking platforms facilitate real-time global communication inclusive of photos and videos, with friends, family, colleagues, and businesses. The initiation and sustenance of interpersonal relationships have noticeably been transformed, stimulated by social media platforms.

This novel texting approach engendered the concept of Internet Linguistics, a distinct study area. Typographical mistakes or typos are common in CMCs [Kyongho Min and William H. Wilson 2000]. Typos can upend natural language processing (NLP) tasks in several ways. Typos can cause the text to become ambiguous, making it difficult for NLP models to figure out the intended meaning. An inaccurate analysis or interpretation might emerge from a single lost letter, or a character being switched, which creates another word with an entirely different meaning. Typos result in words that are not in the dictionary. NLP models are often trained on huge datasets, and they discover the statistical patterns and correlations between words. Typos can produce words that are not found in the training data, which can result in inaccurate predictions or misunderstandings. The names of individuals, locations, organizations, and other named entities can be distorted by typos. This can impede tasks like named entity recognition and entity linking etc., Typos can result in missed matches or inaccurate search results in tasks involving information retrieval or search engines. Typos can cause errors in translation while translating text between languages or make it more challenging for the model to produce appropriate results. Even a small misspelling in the original text can lead to serious mistakes in the translation. Typos may jeopardize the flow of created content in tasks like text creation or summarization and lead to statements that are illogical or grammatically incorrect. The tone of a sentence can be changed by a typo. For instance, a typo may lead someone to interpret a positive text as negative because the mistake resulted in the creation of a negative word. Typos make the wrong sentimental analysis. Typos mess up a sentence's grammatical construction, making it difficult for dependency parsers to accurately determine the links between words. Spoken errors can result in inaccurate transcriptions in automated speech recognition (ASR) tasks, which can have consequences for later NLP activities that depend on accurate transcriptions.

This study aims to identify typos observed in Malayalam social media and explore Malayalam social media language from the perspective of online linguistics. It also offers an automatic typo detector and corrector. Malayalam is the spoken as well as the official language of the state of Kerala and in the union territories of Puducherry and Lakshadweep. Malayalam is one of the official languages of the Indian Union. It belongs to the South Dravidian language family. In 2013, the Government of India conferred classical language status for the Malayalam language. The Dravidian family's distinctive phonological and morphological traits are still present in the Malayalam language, which also incorporates numerous Sanskrit linguistic components. Based on geographic locations, cast, and sub-caste systems, the major language Malayalam has a wide variety of dialects and sub-dialects. These factors lead to the feature of diglossia [Faido Marudut et al. 2019], a clear distinction between the formal, literary form of the language and its colloquial forms of speech. Malayalam social media comments (MLSMCs) are living examples of the diglossic [Faido Marudut et al. 2019] feature of the Malayalam language.

The impact of the internet and the continued usage of communication tools has resulted in substantial changes to the Malayalam language. Computer-mediated communication (CMC) modified natural language in several ways, including orthography, graphology, lexicon, grammar, syntax, pragmatics, and style [Martin Mares 2016, OLOJEDE et al. 2018]. “Internese” is the widely used new language fashion on the internet "Internese" is a phenomenon that occurs everywhere and is noticeable in all facets of Malayalam communication. Both the standard and spoken forms of the language exhibit strikingly distinct aspects that indicate the influence of “Internese”.

The people of Kerala have long profited from globalization and advancements in technology. As a result of these changes, CMC is now extensively used by Keralites [Carley, 2016]. Many cultures, languages, and lifestyles from around the world are embraced by the people of Kerala. It has been visible in the Malayalam social media comments' style, semantics, syntax, and vocabulary. Malayalam social media comments (MLSMCs) serve as an excellent illustration of the unique characteristics of social media language or “Internese". Malayalam speakers on social media leave comments in their chosen dialects of the language. To draw greater attention to their comments on social media, users use catchy new terminology and phrases. In doing so, they ignore the significance of the linguistic characteristics of the Malayalam language. In social media comments, most users place more emphasis on colloquial language. In colloquial language, the speakers ignore the morphotactics of words. The user’s ignorance of the morphotactics of

words leads to various types of errors from the word level to the sentential level. The correct morphosyntax of a lexical element plays a vital role in determining the grammatical functions of words. A simple typographical error causes to frame of ungrammatical sentences. This study does not examine all the factors behind typographical errors or typos made by social media commentators.

A thorough analysis of this study's collected social media comments identified well over 25 types of typos in MLSMCs. Different types of typos like intentional typos, typos due to the flipping of letters, typos due to the omission of letters, and typos due to the addition of letters, etc., are observed in social media comments.[1]. Typing mistakes occur when writing in Malayalam due to the distinctive features of the language and the difficulties created by the restricted Unicode support for Malayalam phonetic elements.

This study approaches the automatic typo detection and correction of Malayalam social media comments in two different methods. At this study's initial stage, we developed a deep-learning (DL)technique employing a sequence-to-sequence (Seq-2-Seq) model. The outcomes produced by this model need to meet the desired quality standard. The complexity of the Malayalam phonograms and the model's excessive reliance on data have decreased its efficacy. This study then looked at the linguistic nuances related to Malayalam phonograms and developed a phonogram-based hybrid model. It was clear from comparing the results of the two models that the hybrid model generated a more accurate result. Increasing the system's data size can improve the performance of the Seq-2-Seq DL model.

2. Literature review

The history of research on automatic spell detectors and correctors is quite long. Different research groups from various natural languages produced considerable research and developed different types of automatic spell detectors and automatic spell correctors using their methods. The Automatic detection of typos, followed by correction, are essential components of conventional NLP systems, machine translation, and Q&A systems. Research communities have been active in pursuit of diverse techniques like rule-based systems, statistical methods, supervised machine learning techniques like Naïve Bayes and support vector machines with labelled data sets, neural networks like Long short memory (LSTM) or transformer models, edit distance algorithms in this method, most researchers followed Levenshtein distance, N-gram models, Contextual language models, and combination approaches.

[Lehal, 2007] developed a Punjabi spell checker. The system's dictionary determines the final output of the system. [Grobbelaar and Kinyua, 2009]. He proposed an automatic spell checker for the African language, Sesotho (aka Southern Sotho), with the help of several algorithms. This system uses a subset of the lexicon, and if a specific misspelled word occurs, it is added to the system’s subset and used to generate the list of suggestions. This subset is called the “short list" and consists of the words whose lengths are within +/- 2 units of the length of the misspelled word. [Islam and Inkpen, 2011] proposed an unsupervised method to determine typographical and grammatical errors using a similarity function and a frequency value for the Bangla language. [Baba and Suzuki, 2012] developed a Keystroke Log Analysis method for the English-Japanese language pair. They extracted only those words that included the backspace key, then recovered the strings before, and after correction, in two different steps.

[Rao et al., 2012] developed a morphological analyzer-based automatic spell checker for the Telugu language. [Shaalan et al., 2012] applied the morphological generation method for typo detection and correction in Arabic. For this system, they used a list of Arabic words using ‘AraComLexis’ as the source. For the generation of vocabulary, they use a finite-state word generation method. In this system, misspelled words are detected in two ways: direct detection and language model detection. They are using an optimized ranking algorithm to predict the correct word. [Sharma and Jain, 2013] developed an automatic spell checker for the Hindi language with a dictionary lookup. They treated only two types of errors: non-word errors and real-word errors. For non-word errors, they calculated strings at edit distances one and two from the identified erroneous string and filtered out words that were not present in the dictionary. They used the Damerau-Levenshtein edit distance method. It yields a set of words that are potential corrections for the erroneous word. For real-world -errors, they created 2-grams and 3-grams along with their frequencies of occurrence. Every 2-gram, 3-gram, and 4-gram of the input text was cross-verified with a collection of vocabulary developed by the computer to detect real-world errors. If the gram frequency is low, it is considered a

typo. To produce corrections for the erroneous gram, they calculated edits of each word in the gram, followed by the construction of valid candidate grams from these.

In 2013 Cordeiro de Amorim and Zampieri developed an automatic spell checker for the English language. Here, they were using a clustering algorithm for the English language. [Basri et al., 2013] proposed a system for identifying misspelled Malay words. Their system possessed a tokenizer, irrelevant symbol remover, English word eliminator, stemmer, and misspelled word identifier. Generally, in Malaya, the spoken forms of words are not typed according to the standard spelling rules, so the spoken forms of the words are considered typos in Basri's system. This system can be utilized only for the classic forms of the Malaya words. Words. In 2014, Kaur and Garg developed a hybrid system for spell-checking and Grammar-checking for the Punjabi language. This Punjabi Spell-checker followed traditional methods like dictionary lookup This Punjabi Spell-checker followed traditional methods like dictionary lookup. The correct word Automatically corrected words are passed through a morphological analyzer and POS tagger to correct the grammar of the input sentence. In 2014, Xin developed an improved graph model automatic spell checker for the Chinese language. The system builds a directed acyclic graph (DAG) from the input sentence for error detection, and this graph model can consider the sentence whole but not a single character.

In 2015, Kengatharaiyer and Segar developed a contextual spell checker for the Tamil language. They proposed a method to correct the contextual spelling mistakes due to the characters in the confusion set. The phonemic similarities of the members of *'l-class' as /la/,/ ɭa/ and / ɻa/ 'ன -class' as / n̪/,/ n̪4th /,/ ɳ/ and 'r-class as /r5th/, and /r/* considered as the confusion set. This system has two modules. The first module detects errors by a dictionary lookup, and the second module corrects the spelling mistakes with an N-gram module. [Choudhury et al., 2018] developed a context-sensitive spell checker for the Assamese language. This system can correct only real-world typos. This system works with the principles of an n-gram language model. [Jin, 2019] proposed an integrated Parts-Of-Speech-tagging error detection system for the Korean language that utilized two schemes. The first method used the Python package Detection of Errors and Correction in Corpus Annotation (DECCA). The second method entailed multiple taggers' disagreement, implemented as a comparison algorithm using each tagger's tagging results. [QI LV et al., 2022] developed a competitive general speller named ECSpellfor Chinese. In this system, they used unlabelled domain-specific data sets, deriving pre-training data sets via an "Error Consistent Masking Strategy." This masking strategy first specifies an error type of the given sentence, then masks single or continuous characters according to inject spelling errors. These error sets are applied to large-scale corpus for unsupervised machine learning. The error-consistent masking strategy involves many steps. This masking strategy first specifies an error type of the given sentence, then masks single or continuous characters according to inject spelling errors. These error sets are applied to large-scale corpus for unsupervised machine learning. The error-consistent masking strategy involves many steps. [Cissé andzFatiha Sadat 2023] developed a non-word error detector and corrector for the Wolof language. They proposed a hybrid approach with tire data structure, dynamic programming, and weighted Levenshtein distance algorithms.

3. Data Preparation

To get Malayalam comments, we've selected a few of the most popular channels from a variety of categories, including news, entertainment, and vlogs. We use the YouTube Application Programming Interface (API) version 3 (v3) from Google for our data extraction. We've collected comments from 11 well-known Malayalam YouTube channels, with an average of close to 5 million subscribers. We have collected 1,312,479 words in total from different media sources. We have collected 1,312,479 words in total from different media sources. Based on the scripts and emoticons, we manually divided the entire collection of comments into various groups. The various comment categories and total number of comments are listed in Table 1.

Comments-types	Number of comments
comments with Emoticons only	138034
comments with Roman Script only	356921
comments with Malayalam Script only	116905
comments with Emoticons in Roman Script	454564
comments with Emoticons in Malayalam Script	128987
comments with Emoticons in both Roman and Malayalam Script	40775
comments with Emoticons in both Roman and Malayalam Script	40775
comments in Roman and Malayalam Script	72596
comments with Unknown	3697

Table 1. MLMSCs categories

Before being corpora, the collected social media comments go through several pre-processing stages. The first stage of pre-processing is removing emoticons from the comments. Some of the emoticons and Malayalam script characters with similar Unicode escape sequences get removed from the dataset, which is one of the major issues found while doing the cleaning. To avoid this situation, we introduce an emoji library that converts Unicode emoticons to Common Locale Data Repository (CLDR) short names and uses regex to remove the short names. After cleaning the data sets, the rows with empty or non-existing values are removed from the corpus. Removal of the Roman script from the dataset is the next pre-processing step. Through this, we obtained comments with Malayalam scripts only. White space is used as a delimiter for tokenizing the words from the sentence. We developed a vocabulary for the system from the tokenized words. This vocabulary is manually filtered and tagged as correct words and misspelled words. The detected misspelled words were replaced with their correct form. After that, using phonogram tokens, the remaining words are tokenized. Then, the machine created a vocabulary of 10,000 unique words for the Malayalam language, and 252 typo words were randomly selected. Statistics of unique words built from the collected corpus is shown in figure 1.



Fig. 1. Statistics of unique words built from the collected corpus.

4. Experiments and discussion

This study proposes two methods for the automatic typo detector and automatic typo corrector for the Malayalam language. First, we attempt a custom Bi-directional Seq-2-Seq model. And we then tried to develop a hybrid model. In both models, we considered the phonogram system of the Malayalam language.

4.1. *Sequence-to-sequence (Seq2Seq) model*

NLP uses a neural network design called a sequence-to-sequence (Seq2Seq) model for tasks like spell-checking. The two primary parts of it are an encoder, which converts input text into a fixed-size representation, and a decoder, which creates corrected output sequences. During the training, the Input-output pairs are aligned. Through this, the spell-

checking model gains the ability to comprehend the context and offer precise word substitutions for misspelled words.

Seq2Seq models are pivotal in enhancing text precision and quality. Specifically, the Seq2Seq model deployed for Malayalam language spell-checking and correction employs a sophisticated approach with custom features. This model is distinct in its use of a bi-directional framework, integrating a tailored encoder and decoder operating at the phonogram level. Such a design permits an in-depth analysis of the phonetic aspects of words. The encoder comprehensively grasps phonograms in both forward and reverse orders, enriching the understanding of context and inter-phonogram relationships. The decoder then utilizes this context vector to generate a corrected sequence. This innovative architecture enables the Seq2Seq model to accurately identify and amend spelling errors, proving exceptionally useful for the Malayalam language's complex spell-checking and correction tasks. Workflow of the proposed Seq2Seq model is shown in figure 2.

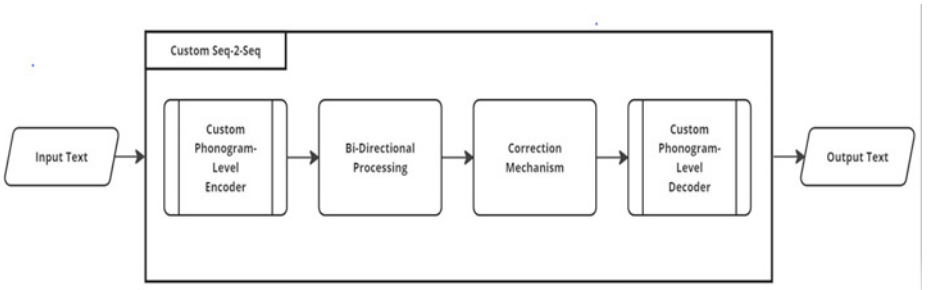


Fig. 2. Workflow of the proposed Seq2Seq model

However, the Seq2Seq model's effectiveness in handling Malayalam's spell-checking challenges is somewhat limited by its dependency on extensive training data. This reliance can reduce its efficiency in dealing with the language's intricate rules, including unique characteristics and exceptions.

4.2. *Hybrid model*

The system combines a rule-based approach, graph, and language models. The system first checks to see if all lexical items match up with its vocabulary before accepting input from a new sentence. A hash table is created based on the phonogram to decrease the time complexity of the lookup. If the word is present in the vocabulary, it indicates that there is no typographical error present in it. Suppose the word is absent from the defined vocabulary, tagged as a typo, and passed a thorough rule-based tokenizer. This tokenizer separates or tokenizes groups of phonemes into valid phonograms. For example: പ്രവീണ /praʋiːɳa] becomes
‘പ്ര’[pra],’ വ്’ʋi:], ‘ണ’[ɳa]. Usually, this word is phoneme tokenized to *[‘പ്ര’,[p] ‘ ്’, ‘വ’[v] ‘ ീ’, ‘ണ’ ɳa]*.
But this kind of tokenization makes it hard to rectify the typo. The phonogram tokens are passed to the language model, which consists of unigram, bigram, and trigram probability mapping. Here, a probability score for each token is calculated. A bidirectional approach calculates the score for each phonogram considered the word ‘പൊതുജനങ്ങളുടെ’[poṯuɻanaŋŋaluṯe] (for the people). Tokens are: ['പൊ',*[po]* 'തു'*[ṯu]*, 'ജ'[ɻa*],* 'ന'*[*na*]*, 'ങ്ങ'[ŋŋa], 'ള്',[lu] 'ടെ'[ṯe]. Now to calculate the score of the token ' ɻa ' both probabilities of *p*('പൊ',*[po]* 'തു'*[ṯu]*| 'ജ'*[*ɻa*]*) and *p*('ങ്ങ'*[*ŋŋa*]*, 'ന'[na*]* /'ജ'*[*ɻa*]′*) are calculated. It increases the positional accuracy of the phonogram. Modifying a particular phonogram is done by extracting all possible valid possibilities. The graph model consists of phoneme level error mapping that gives all possible replacements for each phoneme in Malayalam. Subgraph plotting phoneme relations of /p/ is shown in figure 3.

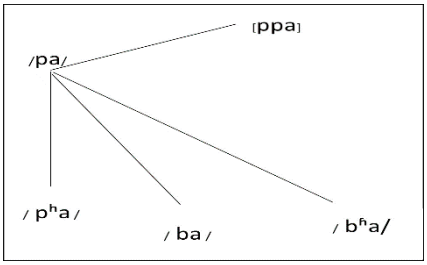


Fig. 3. Subgraph plotting phoneme relations of */p/*

Once the possible phonemes are retrieved, it is replaced in the respective phonogram. All valid phonograms are passed back to the language model, where it is replaced. Generated words are looked up in the vocabulary, and each correctly spelled word is given a probability score. The word with the maximum score is selected as the correct form of the typo. Figure 4 is a self-explanatory of the proposed system.

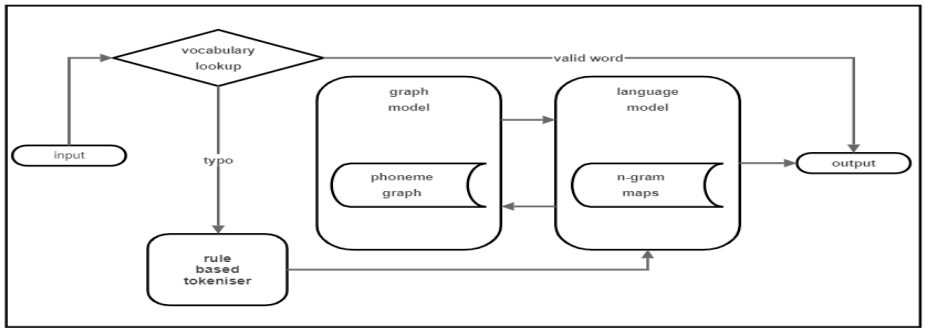


Fig.4 System Architecture of Hybrid Model

4.3. Comparison between the Hybrid system and the custom Seq2Seq model

The Seq2Seq approach, which includes a customized phonogram-level encoder and decoder for Malayalam spell checking, is a convincing option. However, it's essential to address its inherent limitations compared to the hybrid system developed. One of the major limitations of the Seq2Seq model is the significant reliance on the training data. Languages have well-defined linguistic structures that serve as the basis for understanding and analyzing the language. These linguistic patterns imprinted deeply within a language determine the proper word structure and order. The Seq2Seq approach may undervalue the significance of these well-established language principles since it primarily relies on data-driven insights.

The subtle phonetic variations and complex grammatical patterns that characterize languages like Malayalam are es sential. The hybrid approach eliminates the disparity between the flexibility of data-driven models and the rigid character of language rules by smoothly integrating a rule-based mechanism with deep learning components. It successfully combines the strength of deep learning algorithms with the pre-existing linguistic standards, resulting in more precise and contextually aware spell checking and correction. It successfully combines the strength of deep learning algorithms with the pre-existing linguistic standards, resulting in more precise and contextually aware spell checking and correction. Furthermore, due to its high reliance on training data, the Seq2Seq model may need to improve when dealing with complex language rules. Although it could be very good at seeing large-scale linguistic patterns, its ability to understand the subtle quirks and exceptions within a language might be limited. In the context

of a language like Malayalam, the Seq2Seq model's inherent limitations are clear, where a minor change in phonograms completely transforms the meaning. The hybrid system, strengthened by a rule-based strategy, accepts these subtleties. The hybrid system's ability to fully understand the intricate structure of languages like Malayalam is made possible by the mix of linguistic norms and data-driven learning. The hybrid system offers a comprehensive and superior method for spell checking and correction by recognizing pre-existing language norms and fusing them with the benefits of contemporary deep learning techniques. Consequently, compared to the Seq2Seq model's pure deep learning method, the hybrid system regularly outperforms it. Its capacity to consider language conventions, account for exceptions, and use data-driven insights results in spell-checking and correction that are more precise and trustworthy. While the Seq2Seq model with phonogram-level processing shows promise, its inability to adapt to complex linguistic rules highlights the benefits of the hybrid system, which will eventually lead to improved spell-checking and correction skills.

5. Result and Analysis

The phoneme analysis of phonograms significantly enhanced our typo correction techniques. The proposed hybrid model shows an accuracy of 91.26 % with a vocabulary of 6,000 words. The morphosyntax of certain Malayalam words doesn't come under the generalization done by the system. So that kind of lexical item is not processed by the NLP machine. The lack of Unicode for clusters and certain phonograms is one of the obstacles in Malayalam Computing. Developing a full-fledged typographical error detector and rectifier for the Malayalam language needs one whole phonogram representation in Unicode. Along with this, one NLP machine must build with exact knowledge of the morpho-syntax of the Malayalam words to address the issue from its top to bottom levels. Performance analysis of the Hybrid-model is shown in figure 5.

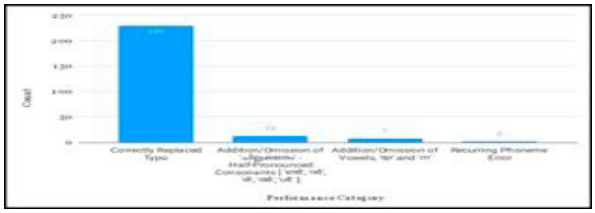


Fig. 5. Performance analysis of the Hybrid-model

6. Discussion and Conclusion

The ubiquitous nature of Computer-Mediated Communication (CMC) has transformed human interactions, fostering a novel culture of internet communication globally. This study highlights the significant impact of CMC on language, with a focus on the emergence of "Internese", a new linguistic style adapted for CMC in Malayalam social media comments.

"Internese" poses unique challenges to numerous Natural Language Processing (NLP) tasks. This linguistic adaptation, reflecting a blend of traditional language elements with digital communication nuances, necessitates a fresh perspective in linguistic study, notably in the domain of internet linguistics. Typographical errors, commonly known as "typos," present significant challenges in NLP, as they can drastically affect the interpretation and processing of language data. In this context, the study focuses on Malayalam social media comments (MLSMCs). The creation of corpora, vocabulary, and language models from MLSMCs forms the crux of this research, aiming to effectively identify and correct typographical errors prevalent in Malayalam social media discourse.

This study, centring on the development of a phonogram-based approach for auto-typo correction in Malayalam social media comments, marks a significant step forward in computational linguistics for Indic languages. The introduction of two models—a deep learning-based sequence-to-sequence (Seq2Seq) model and a hybrid model—reflects a nuanced understanding of the Malayalam language's unique phonetic structure. This approach not only addresses the prevalent issue of typographical errors in social media comments but also opens new avenues in language technology for Malayalam.

The study proposed a hybrid and Seq2Seq model for the auto-typo detector and rectifier for the Malayalam language with a phonogram approach.

The hybrid model under consideration addresses the specific phonological and phonogram characteristics of the Malayalam language. Its performance, evaluated using a newly developed corpus, reached satisfactory levels. While the Seq2Seq model shows potential for Malayalam spell-checking, it encounters difficulties in adapting to the language's complex linguistic nuances. Its dependency on extensive training data limits its effectiveness against intricate linguistic phenomena, including unique language features and exceptions.

In contrast, the hybrid model combines phonogram analysis with rule-based techniques, demonstrating notable efficacy in identifying and rectifying typographical errors, especially those prevalent in colloquial Malayalam as used on social media. This model's capacity to accommodate the language's complexity and variability distinguishes it from the Seq2Seq model. The latter, despite its promise, is constrained by its reliance on large datasets and struggles with the intricate nature of Malayalam.

By compiling corpora and developing language models from MLSMCs, the study provides valuable insights and practical solutions for addressing typographical errors in this digital communication context. The exploration of Malayalam, a language rich in phonetic and morphological complexity, in the sphere of internet linguistics, not only enriches the field but also sets a precedent for future research in other languages influenced by digital communication trends. The methodologies and findings of this study contribute significantly to the advancement of NLP techniques, particularly in addressing the unique challenges posed by "Internese" and its impact on language processing.

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